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**Group 4**

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**SmartSpend AI**  
**Software Development Plan**  
Version 1.0

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| SmartSpend AI             | Version: 1.0   |
| Software Development Plan | Date: 02/07/26 |
| SS-SDP-01                 |                |

## Revision History

| Date     | Version | Description | Author               |
|----------|---------|-------------|----------------------|
| 02/07/26 | 1.0     | Draft       | Nguyễn Trần Lan Viên |
|          |         |             |                      |
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# Software Development Plan

## 1. Introduction

This document outlines the Software Development Plan for SmartSpend AI, a smart personal finance management application integrated with Artificial Intelligence. It defines the project scope, team organization, schedule across 5 sprints, and risk management strategies.

## 2. Project Overview

### 2.1 Project Purpose, Scope, and Objectives

SmartSpend AI is an AI-powered personal finance assistant designed to completely automate transaction logging, addressing the critical drop-out rate associated with tedious manual entries in traditional apps. The application enables users to capture invoices or bank transfer screenshots, from which an AI engine automatically extracts text, classifies financial categories, and delivers tailored financial feedback. The core objective is to deliver a fully functional, mobile-responsive software system containing 15 to 20 screens.

### 2.2 Assumptions and Constraints

- **Schedule Constraint:** The project is bounded by a strict academic timeline structured into 5 Jira Sprints (12 weeks).
- **Budget Constraint:** This is a zero-budget academic software development project.
- **Staff Constraint:** The team is fixed at 4 members with no external additions throughout the project lifecycle.

### 2.3 Project Deliverables

The project deliverables are strictly mapped to the course assignments and evaluation milestones across 5 sprints, structured as follows:

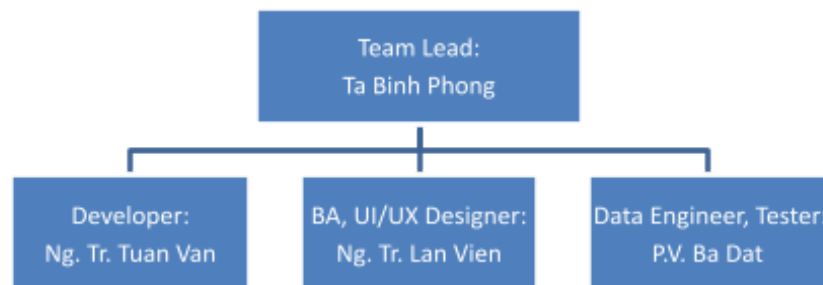
| Assignment | Sprint   | Phase (RUP)  | Key Deliverables & Artifacts                                                                                                                                                                                                                                     | Target Date |
|------------|----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| PA0        | Sprint 1 | Inception    | <ul style="list-style-type: none"> <li>• Software Development Plan (SDP)</li> <li>• Group registration details</li> <li>• Initial Project Proposal with User Interview Evidence</li> </ul>                                                                       | 28/06/2026  |
| PA1 & PA2  | Sprint 2 | Elaboration  | <ul style="list-style-type: none"> <li>• Requirements document (Vision document, Use Cases)</li> <li>• Initial System Design &amp; High-fidelity UI/UX Prototype (Figma)</li> </ul>                                                                              | 12/07/2026  |
| PA3        | Sprint 3 | Elaboration  | <ul style="list-style-type: none"> <li>• Detailed System Design &amp; Software Architecture Document</li> <li>• Initial Test Planning &amp; Test Cases</li> <li>• First Working Software Demo (Core CRUD features)</li> </ul>                                    | 26/07/2026  |
| PA4        | Sprint 4 | Construction | <ul style="list-style-type: none"> <li>• Advanced Working Software iteration</li> <li>• Core AI Modules implementation (OCR Scanner &amp; Auto-Categorization)</li> <li>• Updated Test Reports &amp; QA logs</li> </ul>                                          | 09/08/2026  |
| PA5        | Sprint 5 | Construction | <ul style="list-style-type: none"> <li>• Final Working Software (Fully operational SmartSpend AI application)</li> <li>• Final AI Chatbot integration (Financial Advisor)</li> <li>• Project Status/Weekly Reports &amp; Final Presentation materials</li> </ul> | 23/08/2026  |

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### 3. Project Organization

#### 3.1 Organizational Structure

The SmartSpend AI project team is structured as a cross-functional Agile unit where roles are assigned based on individual specializations to support iterative deliveries across 5 sprints. Project management and oversight are primarily steered by the Project Manager, who facilitates team synchronization via JIRA and Slack/ Discord.



#### 3.2 Roles and Responsibilities

| Person                | Role                     | Responsibilities                                                                         |
|-----------------------|--------------------------|------------------------------------------------------------------------------------------|
| Tạ Bình Phong         | Project Manager (PM)     | Project planning, Jira & Sprint management, milestone tracking, OCR/LLM API integration. |
| Phan Văn Bá Đạt       | Data Engineer / Tester   | Database design, transaction data processing, writing test cases & QA.                   |
| Nguyễn Trần Lan Viên  | UI/UX Designer / Report  | High-fidelity wireframes design, writing proposals & reports.                            |
| Nguyễn Trịnh Tuấn Văn | Software Developer (Dev) | Frontend & Backend development.                                                          |

### 4. Management Process

#### 4.1 Project Estimates

- Estimated Cost: Zero-budget.
- Estimated Schedule: 10 weeks in total, structurally divided into 5 sprints.
- Basis for Estimates: The milestone allocation is strictly aligned with the predefined syllabus roadmap and deadlines of the Introduction to Software Engineering course.
- Re-estimation Points: The project team will reassess and re-estimate task granularities at the end of Sprint 2 (Elaboration Phase), immediately following the completion of finalized database schemas and high-fidelity UI wireframes.

#### 4.2 Project Plan

##### 4.2.1 Phase and Iteration Plan

The project lifecycles follow a hybrid framework integrating both RUP (Rational Unified Process) phases and Agile Scrum sprints.

- Phase 1: Inception
  - Overall Objectives: Conduct preliminary user research, complete team assignment configurations, and establish the initial project proposal (PA0).
  - Iteration Tasks:
    - + Formulate the core product concept of the "SmartSpend AI" application.
    - + Interview at least 5 potential users to extract real-world pain points and log qualitative evidence.
    - + Author and deliver the baseline Software Development Plan (SDP).

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- Phase 2: Elaboration
  - Overall Objectives: Finalize requirements extraction, architectural specifications, comprehensive test planning, and deliver a basic working CRUD prototype (PA1, PA2, PA3).
  - Iteration Tasks:
    - + Sprint 2 (29/06 – 12/07): Formulate the Project Vision document and define foundational functional Use Case diagrams. Construct high-fidelity UI/UX layouts in Figma.
    - + Sprint 3 (13/07 – 26/07): Draft programmatic UML Sequence and Class diagrams; formulate software test plans; develop the first working prototype executing basic CRUD processes (Authentication, Profile settings, Dashboard charts, Manual Entry).
- Phase 3: Construction
  - Overall Objectives: Implement advanced intelligent automation engines, complete comprehensive integration testing, and finalize product delivery (PA4, PA5).
  - Iteration Tasks:
    - + Sprint 4 (27/07 – 09/08): Integrate the AI Smart Scanner leveraging external Computer Vision (OCR API) and deploy the automated transaction category tagging system.
    - + Sprint 5 (10/08 – 23/08): Embed the AI Financial Advisor conversational chatbot; conduct intensive system quality testing; resolve active bugs and pack the final operational deliverables.

#### 4.2.2 Releases

- Release v0.1 (End of Sprint 3 - 26/07): Alpha Version - A functional software prototype executing core CRUD features without intelligent automation integrations.
- Release v1.0 (End of Sprint 5 - 23/08): Final Production Version - The fully integrated SmartSpend AI platform providing seamless automated receipt parsing, semantic categorizations, and reactive chatbot features ready for grading.

#### 4.2.3 Project Schedule

| Phase (RUP)  | Sprint (Scrum) | Start | End  | Assignme nt | Main objectives                                      |
|--------------|----------------|-------|------|-------------|------------------------------------------------------|
| Inception    | 1              | 14/6  | 28/6 | PA0         | Project proposal                                     |
| Elaboration  | 2              | 29/6  | 12/7 | PA1, PA2    | Requirements and initial design.                     |
| Elaboration  | 3              | 13/7  | 26/7 | PA3         | Detailed design and test planning. Working software. |
| Construction | 4              | 27/7  | 9/8  | PA4         | Working software. Implementation and testing         |
| Construction | 5              | 10/8  | 23/8 | PA5         | Working software. Implementation and testing         |

### 4.3 Project Monitoring and Control

#### 4.3.1 Reporting

To ensure transparent progress alignment and continuous tracking across the 5 sprints, the team implements the following multi-tiered reporting structure:

- Weekly Meetings: Conducted offline every Friday morning to audit the JIRA sprint backlog, review active task state changes, and finalize upcoming assignments.
- Weekly Status Report: The Project Manager compiles a structured progress summary report at the end of each sprint cycle.
- Informal Chats: Managed continuously through the group's synchronized Slack/ Discord workspace channel.

#### 4.3.2 Risk Management

*[Identify risks in your project. The risks should be prioritized, and shorted according to their priority.]*

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| <i>Risk ID</i> | <i>Risk Description</i> | <i>Probability</i> | <i>Impact</i> | <i>Risk Exposure</i>  | <i>Priority</i> | <i>Mitigation Strategy or Contingency Plan</i> |
|----------------|-------------------------|--------------------|---------------|-----------------------|-----------------|------------------------------------------------|
|                |                         |                    |               | =Probability * Impact |                 |                                                |
|                |                         |                    |               |                       |                 |                                                |
|                |                         |                    |               |                       |                 |                                                |

4.3.3 Configuration Management

- Document and file: are stored and shared on Google Drive.
- Source Control and Version Management: are hosted via GitHub under the strict structural folders requested.